Utilization and Profitability of Tractor Services for Maize Farming in Ejura-Sekyedumase Municipality, Ghana

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ABSTRACT

Maize farming is a major livelihood activity for many farmers in Ghana. Unfortunately, farmers usually do not obtain the expected returns on their investment due to reliance on rudimentary, labour-intensive, and inefficient methods of production. Using cross-sectional data from 359 maize farmers, this study investigates the profitability and determinants of the use of tractor services for maize production in Ejura-Sekyedumase in the Ashanti region of Ghana. Results from descriptive and profitability analyses reveal that tractor services such as ploughing and shelling are widely used, but their profitability varies significantly among farmers. Key factors influencing profitability include farm size, fertilizer quantity applied, and farmer experience. Results from a multivariate probit analysis also showed that farming experience, fertilizer quantity, and profit per acre have a positive influence on tractor service use for shelling, while household size, farm size, and FBO membership have a negative influence. Farming experience, fertilizer quantity, and profit per acre positively influence tractor service use for ploughing, while farm size has a negative influence. A t-test result reveals a statistically significant difference in profit between farmers who use tractor services and those who do not. Specifically, farmers who utilize tractor services on their maize farm had a return to cost of 9% more than those who do not ($p < 0.05$). Kendall's result showed a moderate agreement among the maize farmers, with financial issues being the most ranked constraint in accessing tractor services. However, the study is limited to the perspectives of maize farmers, excluding insights from tractor service providers and policymakers that could enrich the analysis. It also does not consider climatic and environmental factors that influence maize profitability. Future research should include these additional stakeholders and variables, and consider using a balanced scorecard approach to evaluate both financial and non-financial aspects of tractor service performance. The study recommends that the Ministry of Food and Agriculture and other stakeholders should provide financial assistance to enable farmers to access tractor services, strengthen extension services to guide best practices, and promote the use of quality seeds and fertilizers to improve maize farm profitability

Keywords: Maize Farming; tractor Services; Utilization; Profitability; Ejura-Sekyedumase.

Introduction

Agriculture is not only a vital sector in Ghana but also a significant contributor to the country's economy. It provides employment opportunity for large portion of the population, especially rural areas. Additionally, agriculture ensures food security by producing a substantial portion of the country's food needs. Maize (Zea mays) rank first as Ghana's most important cereal produced and consumed. Maize is very important staple cereal product in Ghana, accounting for more than 50% of the total cereal production in the country and grown in all agro-ecological zones (Akramov and Malek 2012). MoFA (2007) characterizes the agricultural sector in Ghana as predominantly practiced by smallholder farmers and family operated farms using rudimentary technology to produce about 80% of Ghana's total agricultural output. Also, over 70% of Ghana's maize production is by smallholder farmers who do not have access to required resources needed for increasing productivity, making them prone to recording low yields (SARI, 1996). Musungwini (2018) notes that smallholder agriculture needs to be viable, productive, and sustainable to achieve food security through the elimination of hunger and the alleviation of poverty, as considered in the United Nations (UN) Sustainable Development Goals (SDGs). But the ongoing reliance on the traditional methods of farming prevents this realization. Hence, the use of tractor services in agriculture will enhance efficiency and effectiveness, enabling farmers to meet their needs quickly while ensuring continuity and profitability.

Tractor service is not an end in itself, but rather a means to achieve an end, serving as a catalyst in improving productivity, a likely positive effect on farmers' income and overall wellbeing. The history of today's mechanized countries has demonstrated how drastically the face of agricultural and rural areas in Africa would alter if mechanical power were widely substituted for manual labor (Jansen, 1969). Since the late 1980s, Ghana's economy has experienced steady growth, accompanied by increased non-farm opportunities in rural areas and rapid urbanization. During this period, the government has adopted a market-driven approach, avoiding sector-specific favoritism, except in the cocoa industry. However, in recent years, the government has shifted focus towards agricultural mechanization,

directly importing tractors, establishing subsidized service centers, and implementing policies to promote sustainable growth and development in the agricultural sector.

Currently there are limited studies focusing on maize production in Ejura-Sekyedumase Municipality that specifically examine the profitability of maize farmers in the area. Notable examples of existing research include "Development of Agricultural Mechanization in Ghana" (Cossar *et al.*, 2016), "Status of Agricultural Mechanization in Ghana" (Taiwo and Kumi, 2015), "The Effect of Climate Variability on Maize Production in the Ejura-Sekyedumase Municipality, Ghana" (Codjoe et al., 2021), "The 2020 Maize Production Failure in Ghana" (Obour *et al*., 2022), "Assessment of the Youth in Agriculture Programme in Ejura-Sekyedumase District" (Baah, 2014), and "Assessment of Tractor Maintenance Practices of Tractor Operators at Ejura, Ghana" (Aikins *et al*., 2016). However, there is a notable gap in studies that specifically investigate the profitability of maize farmers who employ mechanization (tractor services) for maize farming within this municipality. This gap is particularly concerning given the increasing emphasis on mechanization (tractor services) as a means to enhance agricultural productivity and economic returns. While previous studies have explored various aspects of maize production, such as yield potential and technical efficiency, they often overlook the financial implications of adopting tractor services. Understanding how tractor services impact profitability is crucial for farmers who are considering investments in such services. Moreover, without empirical data on the profitability associated with tractor services, policymakers and agricultural stakeholders may struggle to formulate effective strategies that encourage mechanization among maize farmers. Therefore, this study aims to examine the profitability and determinants of the utilization of tractor services for maize production in Ejura-Sekyedumase Municipality in the Ashanti region of Ghana. By addressing this critical gap, the research will provide valuable insights that can inform both farmers' decisions and policy interventions aimed at promoting mechanized agriculture in Ejura-Sekyedumase Municipality.

The research aims to pinpoint constraints within the tractor services industry, providing essential insights and empirical data that could boost agricultural productivity in the municipality. The findings are intended to motivate potential investors and stakeholders to leverage the results for informed decisions about utilizing tractor services effectively. Furthermore, these insights will support policymakers in formulating appropriate policies that enhance access to tractor services for local farmers. Moreover, the study will also contribute to the body of literature on agricultural mechanization (tractor service), potentially serving as a basis for future research.

Literature Review

Tractor Services Utilization

Agricultural mechanization, particularly through tractor services, can be defined as the economic application of machinery to enhance the efficiency and productivity of human labor in farming. Tractor services serve not merely as an end goal but as a vital means to accelerate agricultural productivity. The adoption of mechanized agriculture has the potential to be extremely significant as it will assist in mitigating most of the challenges faced by smaller farmers. This includes enhancing agricultural output, promoting sustainable development in many Ghanaian rural communities, and improving living conditions. The implementation of tractor service in agriculture holds great potential as it will help to alleviate most of the difficulties that smaller farmers face. This entails raising agricultural productivity, encouraging sustainable development in most rural Ghanaian communities, increasing agricultural productivity, and enhancing quality of life. The mechanization of agriculture represents a significant advancement in the field, as highlighted by Yohanna (2004). Kepner *et al.*, (1978) attribute the substantial rise in agricultural production to the increased use of non-human energy sources and improved machinery. The widespread adoption of tractor services in developed nations during the 20th century has been recognized as a key factor in achieving higher living standards (Reid *et al.*, 2003).

For developing countries, however, it is crucial to emphasize small-scale farming, which constitutes the majority of agricultural production, to ensure economic stability (Hoki *et al.*, 1992; Yohanna, 2004).

Much of the world's poverty is concentrated in rural areas of developing countries where agriculture remains the backbone of livelihoods, providing both employment and food security (Alston & Pardey, 2014; de Brauw & Bulte, 2021; Perkins *et al.*, 2013). For overall economic growth, it is imperative to prioritize agricultural development, as agriculture plays a substantial role in these economies. Mechanical power technology represents the highest level of agricultural mechanization, encompassing all machinery that derives its primary power from sources other than human muscle. Utilizing tractor services for agricultural operations can significantly reduce production costs by minimizing labor expenses. The benefits of tractor services include increased agricultural productivity and improved living standards for farmers. Nonetheless, concerns have been raised regarding farm mechanization's negative impacts, such as soil degradation and deforestation.

Impact of Tractor Services in Sub-Saharan Africa (SSA)

Agriculture is a vital contributor to the economy of sub-Saharan Africa, significantly contributing to GDP, job creation, and food security. According to the World Bank, sub-Saharan Africa has a GDP of $1.7 trillion, with agriculture accounting for nearly $300 billion, or about 17% of the economy, employing approximately 282 million people, which represents 51% of the region's workforce (Trade Map - Trade Statistics for International Business Development, Sihlobo et al., 2021). A large portion of the population in sub-Saharan Africa resides in rural areas and is directly or indirectly dependent on agriculture. The reliance on manual tools such as hand hoes and mattocks for land preparation by smallholder farmers results in low productivity and income while imposing a heavy workload. This manual cultivation leads to high post-harvest losses, reduced competitiveness, low agricultural exports, and increased imports. Therefore, adopting tractor services could significantly improve these conditions. The importance of mechanization is underscored by the African Union's "Agenda 2063: The Africa We Want," which aims to eliminate the use of manual tools like the mattock by 2025 as part of a broader strategy for agricultural modernization. By transitioning to tractor services, farmers can enhance their productivity and efficiency, ultimately contributing to improved food security and economic stability in the country.

Agricultural mechanization in Africa remains significantly lower than in other continents, with approximately 70% of farmers cultivating small plots of land—typically less than two hectares—using hand hoes (Pingali, 2007). This indicates a persistent reliance on manual labor within the agricultural sector. A study by Taiwo and Kumi (2015) revealed that the level of agricultural mechanization among maize producers in the Ejura-Sekyedumase municipality is still at the pedestrian level. Despite their commercial production efforts, nearly all farming operations depend on manual technology. However, the study suggests that enhancing the availability of quality, durable, and efficient mechanization inputs, coupled with effective extension services and supportive policies, is essential for achieving sustainable agricultural production. According to estimates from the Food and Agriculture Organization (FAO), Africa has fewer than two tractors per 1,000 hectares of arable land. In 2012, sub-Saharan Africa averaged only 1.3 tractors per 1,000 hectares of cultivated land, compared to 9.1 tractors in South Asia and approximately 10.4 tractors in Latin America during the same period. These statistics highlight the stark contrast in mechanization levels across regions, with sub-Saharan Africa exhibiting relatively low adoption of mechanized practices. The need for increased mechanization is further emphasized by the African Development Bank's call for prioritizing agricultural mechanization to boost productivity and reduce reliance on manual labor. Mechanization not only enhances efficiency but also plays a crucial role in addressing labor shortages and improving overall agricultural output.

As the agricultural sector evolves and smallholder farmers face various challenges, it becomes crucial to have farm mechanization methods that are specifically tailored to their needs. Smallholder farming requires appropriate and efficient tools and machinery that can help increase productivity, reduce labor-intensive task, and improve overall efficiency. By providing suitable farm mechanization options for smallholders, it helps them overcome challenges and enhances their livelihoods. This is partly due to government intervention. A study by (Mdoda *et al.*, 2022) revealed that agricultural mechanization has dual extensive impact on smallholder agricultural production, However, the study suggests that the government and other stakeholders must develop effective agricultural access to credit by farmers as a strategy for increasing the adoption attitude of agricultural mechanization among farmers. Tractor service has a dominant role in the development of rural communities. It is not an end, but rather a means to achieve an end, serving as a catalyst for increased productivity and relatively the development of the community as well as the country. In a study by Amare and Endalew (2016), they found that agricultural mechanization bears undisputed truth for improving food security, creating employment opportunities, increasing productivity, reducing losses, and promoting gender empowerment while maintaining environmental degradation at lower levels. However, the study suggests that since adoption of mechanization follows stages, it is better to select the one that is most efficient for better productivity.

Theoretical Framework

There have been several theories published by many researchers centered on the utilization and profitability of agricultural mechanization. The following are the most relevant theories in this regard;

The Profit Maximization Theory

This theory was propounded by Adam Smith (2000); Farmers strategically manage their use of tractor services to achieve their financial goal, to enhance their productivity and overall profitability of their farm operations. By applying the profit maximization theory, maize farmers can streamline their tractor services, reduce costs, and boost profitability, ultimately contributing to a more sustainable and productive agricultural sector.

Utility Maximization Theory

This theory was propounded by Jeremy Bentham and John Stuart Mill (1861); Farmers identify their preferences and goals related to the use of tractor services, such as increasing crop yield, reducing labor cost while improving efficiency or enhancing overall profitability. Farmers also evaluate trade-offs between the different tractor services in terms of investment cost, potential increase in productivity.
The Profit Maximization Theory best fits the analysis of tractor services' utilization and profitability, as it aims to optimize resource allocation to maximize financial returns, aligning with the objective of evaluating the economic viability of tractor services.

Conceptual Framework
Based on the research objectives and the review of relevant studies related to the profitability of the use of tractor services, the conceptual framework for this is organized as follows;

The challenges farmers face in accessing tractor services (e.g. financial issues), socioeconomic and demographic factors (e.g. Age, farmer experience, farm size) and the institutional factors (e.g. income, belonging to farmer-based organizations) affect farmer's choice of using tractor services for maize farming which in turns affect their overall maize farm profitability.

Empirical Review
The government of Ghana has been providing subsidized agricultural machinery to private businesses known as Agricultural Mechanization Services Enterprise Centers (AMSEC) since 2007, aiming to enhance tractor-hire services for smallholder farmers. Despite an average annual growth rate of 1.1% in Ghana's rural population from 1990 to 2010 and a 3.9% increase in cultivated land, the demand for mechanization remains high, particularly for land preparation services like ploughing. Recent statistics indicate that mechanized farms allocate approximately 12% of their expenses to land preparation and 6.1% to crop irrigation. This research quantitatively assesses whether the AMSEC model is appealing to private investors, utilizing a firm investment model and recent data. However, findings suggest that despite government efforts to promote private sector-led mechanization, the AMSEC model is unlikely to be a profitable business venture that attracts private investors, even with existing subsidies. The primary constraint identified is the low utilization rate of tractors due to limited operational scale. The research involving 270 farm households supports the assertion made by Pingali, Bigot, and Binswanger (1987) that heavily subsidized mechanization service centers are not a viable policy option globally, including in Ghana. While the study concludes that the development of a tractor rental market is essential for mechanizing agriculture in smallholder-dominated economies like Ghana, it emphasizes that this market's success depends on several factors. These include increased tractor usage through migration between different rainfall zones, the ability to perform multiple tasks with tractors, and the availability of low-cost tractors.
The research titled "The Influence of Tractor Services on Smallholder Agriculture Productivity: Evidence from Mnquma in the Eastern Cape Province" by (Mdoda *et al.,* 2022) utilized a semi-structured survey to gather data from 120 smallholder farmers through a stratified sampling technique. The analysis employed propensity score matching and logistic regression models, revealing that 90% of the farmers used agricultural machinery. The findings indicate that mechanization positively impacts productivity, farm returns, and crop intensity. To enhance farmers' attitudes towards tractor services, the study recommends that the government and relevant stakeholders improve access to agricultural loans and promote innovative technology adoption. Mohamed (2020) examined the impact of mechanized farming on smallholder farmers in Sierra Leone, using a qualitative methodology with 25 respondents. The study found that mechanization increases labor efficiency, boosts economic growth, and enhances food production, thereby contributing to food security. It concludes that adopting mechanized farming practices is crucial for improving food production and security. Additionally, (Nkegbe *et al.,* 2013) explored the technical efficiency of maize production in Northern Ghana using stochastic frontier methodology with a sample of 360 farmers. The study found an average technical efficiency of 74% and concluded that factors such as gender, experience, and agricultural mechanization significantly affect efficiency. It suggests that young farmers can benefit from the expertise of more experienced ones and emphasizes the need for easier access to tractors through Agricultural Mechanization Services Enterprise Centers.

To the best of our knowledge and on the above experienced base review, the relationship between tractor service and the profitability of the smallholder maize famers have not been fully studied. Per the above, the researchers believe that the utilization of tractor services benefits smallholder farmers and increases their returns, but the study does not consider the profitability of these farmers in using these tractor services. Hence, this study is the next logical step to explore the relationship between utilization of tractor services and the profitability of maize farmers in the study area.

Analytical Review

In a study by Amankwah (2021) on the adoption of multiple agricultural technologies and their impact on productivity in rural Nigeria using a multivariate probit model, the researcher used the multivariate probit model because of the categorical and non-mutually exclusive nature of the technologies considered. The statistical model best fit in analyzing one of the objectives of this study "the factors that influence the choice of use of tractor services". These tractor services (ploughing and shelling) are categorical and are not mutually exclusive as some household may utilize or a combination of the services. The multivariate probit model was used to identify the "factors that influence the choice of tractor services." A study by Chitheka et al., (2024) on analysis for income comparison between members and non-members of livunzu cooperative in chikwawa district, Malawi. The study aimed to test the hypothesis that "there is no difference in farm incomes between Livunzu cooperative members and non-members". The study used the t-test to test whether there is a significant difference between the two group. The statistical models best fit in analyzing one of the objectives of this study "effect of utilization of tractor services on maize farm profitability" between those who used tractor services for farming and those who do not.

In a study by Bempomaa (2014) on technical efficiency in maize production in Ejura-Sekyedumase municipality, the researcher used Kendall's coefficient of concordance to analyze one of the objectives of the study. "To identify the constraints to maize production" This model was used because the researcher assessed the degree of agreement (concordance) among the maize farmers. This model best fit in analyzing one of the study's objectives "constraint farmers face in accessing mechanized farm tools." Even though the Spearman's correlation coefficient can also be used for rank variables, it has some disadvantage in relation to this context. It is also used to find the strength and direction between variables but not the degree of agreement (concordance) as we hope for in this study. Also, unlike the Spearman's rank correlation coefficient the Kendall's coefficient of concordance can also handle ties among the ranks given by the judges (in this case the maize farmers) making it a robust in this analysis.

Research methodology

Study Area and Data

The study area is the Ejura-Sekyedumase municipality, located in the Ashanti region of Ghana (Figure 1). Established in 1988 through the decentralization program, it was formed from portions of the former Sekyere and Offinso Districts. The municipality spans approximately 1,782.2 square kilometers (690.781 square miles), making it the fifth largest district in the Ashanti region, accounting for about 7.3% of the region's total land area. It lies between longitudes 1.5°W and 1.39°E and latitudes 7.9°N and 7.36°S. Ejura-Sekyedumase is bordered by the Atebubu-Amantin district to the north, Mampong municipality to the east, Sekyere South district to the south, and Offinso municipal assembly to the west. The northern part of the municipality features flat and undulating terrain with elevations ranging from 150 to 300 meters, which supports mechanized farming practices. The district capital, Ejura, is situated about 106 kilometers from Kumasi, the regional capital. According to the Ghana Statistical Service (2021), the municipality has an estimated population of approximately 137,672 people, with about 85,250 (estimate) of them identified as maize farmers by officials from the Ministry of Food and Agriculture (MoFA) during a recent visit. The district is located in a transitional zone between semi-deciduous forest and guinea savannah and experiences a bimodal rainfall pattern, with annual averages ranging from 1,200 mm to 1,500 mm. The primary rainy season occurs from April to August, while a minor season lasts from August to November.

The municipality was purposively selected because of the dominant use of tractor services for maize production. Eight (8) communities were purposively sampled from a total of 75 communities for the study. A multi-stage sampling technique was employed in the study, with a sample size of 350 individuals in the municipality. A semi-structured quantitative questionnaire was employed to collect empirical data in the study area. The number of respondents in the selected communities in the municipality using proportionate sampling are shown in Table 1 below.

Table 1: Number of Respondents in the Selected Communities

Communities	Estimated Farmers Population	Number of respondents
Aframso	150	26
Dromankuma	600	106
Babaso	150	26
Teacherkrom	100	18
Kasei	300	53
Samari-Nkwanta	75	13
Sekyedumase	400	70
Bayere-Nkwanta	100	18
	Total = 1875	Total = 330

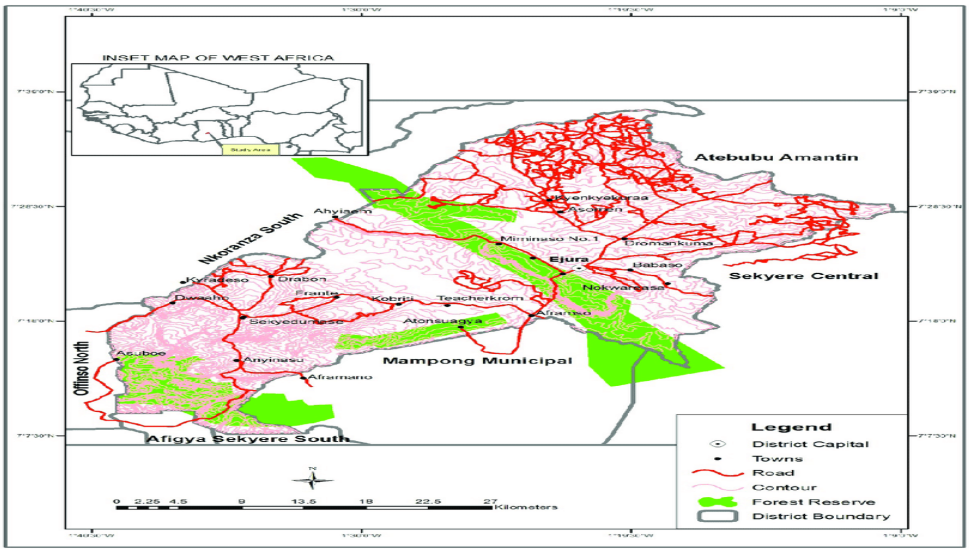

Figure 1: A Map of Ejura-Sekyedumase Municipality
Source: CSIR-Council for Scientific and Industrial Research-Ghana (June 2019).

Data Collection and Quality Assurance

Data was collected through semi-structured face-to-face interviews using a comprehensive 45-item questionnaire that captured quantitative metrics across several domains. The instrument was designed to collect precise numerical data on socioeconomic characteristics, farm production parameters, tractor service utilization patterns, and detailed profitability indicators. Prior to full deployment, the research team conducted extensive pretesting to ensure consistency and reliability. A training and validation protocol was organized to ensure data quality and consistency. Four enumerators underwent intensive three-day training sessions covering standardized question administration techniques, neutral probing methods, and accurate data recording procedures. During the fieldwork phase, the team established multiple quality control checkpoints, including daily verification of completed questionnaires and random follow-up validation of 20% of collected responses. All data underwent dual-entry verification in statistical software packages, with any discrepancies systematically resolved through team consensus to guarantee dataset integrity.

Ethical Framework/Consideration

The study was conducted under strict ethical guidelines designed to protect participant rights and data confidentiality. Before commencing the data collection, enumerators provided comprehensive verbal explanations of the study's purpose, procedures, and intended use of findings, obtaining explicit oral consent from each participant. The consent process emphasized the voluntary nature of participation and participants' right to withdraw at any time without consequence. To safeguard participant privacy, all collected information was immediately anonymized. During analysis and reporting, results were exclusively presented in aggregate form to prevent any potential identification of individual respondents. These measures ensured compliance with ethical research standards while maintaining the scientific validity of the study's findings.

Data Analysis

Using the software Stata-MP 16.0 and the statistical package for social sciences (SPSS), both descriptive and inferential statistics were applied for data analysis. The multivariate probit model was used to analyze factors that influence the choice of tractor services for maize farming in the study area. Having access to tractor service = 1 and otherwise = 0. The multivariate probit model is used for the prediction of the probability of multiple categorical dependent variables that can occur simultaneously. The maximum likelihood estimation was used to capture the relationships between the independent variables and the dependent variables (tractor services).

The general multivariate probit model is specified as follows: $Y^*_{ij} = \beta_j X_{ij} + \mu_{ij,}$ where;

Y^*_{ij} is the latent variable representing the tendency for the *i* individual to use tractor services *j*.

β_j, is the vector of coefficients or parameters for the various tractor services

X_{ij} is the vector of explanatory variables for the *i* individual and tractor services *j*

μ_{ij}, the stochastic error term following a multivariate normal distribution.

Where; $Y^*_{ij} = \{1, \text{ if } Y^*_{ij} > 0 \text{ and } 0, \text{ if } Y^*_{ij} \leq 0\}$

The empirical multivariate probit model is specified as follows:

$$Y(Trac.) = \beta_0 + \beta_1 Fertilizer_quantity + \beta_2 Age + \beta_3 edu + \beta_4 HH_size + \beta_5 Farm_size + \beta_6 Farming\ experience + \beta_7 Profit_per_acre + \beta_8 Fbo + \beta_9 Sex + \mu$$

Where; Y= tractor services, βo=constant, β_1- β_9= coefficient of the independent variables, μ=stochastic error term.

Table 2 provides a description of the independent variables, their measurement and expected sign in relation to the choice of use of tractor services.

Table 2: Factors that influence the Choice of Tractor Services and the Expected sign

Variables	Variables meaning	Measurement	Expected Sign
Dependent variables;			
Trac	Tractor services	Categorical; the various tractor services employed, where (employed=1, not employed=0)	
Independent variables;			
Fertilizer_quantity	Quantity of fertilizer used	Continuous (weight)	+/-
Age	Age of the farmer	Continuous (Years)	+
Education	Years in formal education	Continuous (Years)	+
Sex	Sex of the respondent	Categorical (Male=1, Female=0)	+/-
Farm_size	Farm size	Continuous (acres)	+
Farming_experience	Years in farming	Continuous (in numbers)	+
Profit_per_acre	Profit/acre of maize farmer	Continuous (cedis)	+/-
FBO	Farmer based organization	Categorical; (belong to FBO=1 and otherwise=0	+/-
HH_Size	Household size	Continuous (in numbers)	+/-

To determine the effectiveness of the tractor services in enhancing maize farm profitability, Gross Profit Margin and Net Profit Margin were used. The Gross Profit Margin reveals the disparity between operational costs and revenue, while the Net Profit Margin demonstrates the farmers' true profitability by accounting for all expenses. Using t-test, we compare the mean values of financial indicator (net profit) between farmers who use tractor services and those who do not, to determine if there were significant differences between them.

Gross Profit Margin**:** It is the ratio of gross profit to total revenue expressed as a percentage (Nariswari et al., 2020).

Thus, Gross Profit Margin = $(\frac{Gross\ Profit}{Total\ Revenue}) \times 100$

Total revenue = price per unit × quantity sold

Gross Profit = Total Revenue – Total Variable costs

Net Profit Margin**:** It is the ratio of Net profit to Total Revenue expressed as a percentage (Nariswari et al., 2020).

Thus, Net Profit Margin = $(\frac{Net\ Profit}{Total\ Revenue}) \times 100$

Net Profit = Gross Margin – Total costs

Total costs = Fixed costs + Variable costs

Return to cost =($\frac{Net\ profit}{Total\ cost}$) $\times 100$
Tractor Margin = ($\frac{Tractor\ costs}{Total\ Revenue}$) $\times 100$
Tractor costs = shelling cost + ploughing cost

Statement of Hypothesis
Null Hypothesis (H_0): The use of tractor services has no significant effect on maize farm profitability. Alternate Hypothesis (H_1): The use of tractor services has a significant positive effect on maize farm profitability. Decision Rule: If the p-value < α (0.05), we reject the null hypothesis (H_0) and conclude that the use of tractor services has a significant positive effect on farm profitability. If the p-value ≥ α, we fail to reject the null hypothesis (H_0) and conclude that there is no significant difference in maize farm cost of production between farms that use tractor services and those that do not.

Maize farmers face and rank constraints in assessing tractor services using the Kendall's coefficient of concordance test. The Kendall's coefficient of concordance test is a non-parametric statistical approach used to determine the degree of agreement or concordance among the maize farmers in ranking a set of constraints from most important to the least important. The constraints were ranked using the numbers 1,2,3,4 etc. in that order from most influential to least influential. The sum of each score was calculated and the constraint with the lowest score was determined to be the most pressing constraint while the constraint with the highest score was determined to be the least pressing constraint. The Kendall's coefficient of concordance was computed as follows:
W=$\frac{12\sum d^2}{m^2(n)(n^2-1)}$, $0 \leq W \leq 1$, Where; m = number of raters or judges (in this case the farmers), n = number of items to be ranked, d = Ri – A Ri= the sum of ranks assigned to item (i) by m judges, A = $\frac{\sum R}{n}$

Results and Discussions
Socio-demographic and Production Characteristics of Maize Farmers
Table 3 shows the demographic and socioeconomic characteristics of maize farmers in the study area. The results revealed that 235 maize farmers (representing 65.5%) were males implying that maize farming is highly dominated by the males in the Ejura-Sekyedumase. Most of the respondents (70%) were married. With this information, it indicates that there might be less need for tractor services in their farming operations. The average/mean age of respondents was 45 years which had 18 years and 79 years as the minimum and maximum age respectively. This indicate that maize farming in Ejura-Sekyedumase is dominated by the middle class. With regards to the years in formal education, the farmers have a mean or an average of eight (8) years, which indicates on average the number of years that the maize farmers engage in formal education. The minimum and maximum years in formal education are 0 year and 20 years respectively. These assumes that the higher the years in formal education, the higher their understanding of the benefits and tractor services utilization rate. Out of the 359 respondents, 65.5% were Christians, indicating that most of the respondents were Christians. About 88.6 % had maize production as their main occupation implying maize production as their main source of income and 51% had their own private businesses that they undertake to supplement their income from the main occupation (maize production).

The minimum year(s) of experience in maize farming was one year, with a maximum of 55 years and suggested that the average maize production experience was 16 years. This implies that the maize farmers in Ejura-Sekyedumase have gained enough experiences necessary in the production of maize which is expected to enhance performance. In the aspect of the acquisition of farmland, 53.2% used the leasing method of land acquisition. The minimum number of acres of land for maize production was one, with a maximum of 50, and the average acre of land was 5.84 acres. The more the acres cultivated, the higher the tractor services utilization rate. About 59.6 % were not engaged in any farmers association. This result suggests that either most of the respondents do not know about these farmers' associations or they just decide not to join them. In the context of tractor services utilization, it is assumed that the more a farmer is associated with any association, the better their abilities to utilize the tractor services to increase performance. Most of the maize farmers, about 90% utilize tractor services in their maize production. The use of these tractor services is assumed to increase performance and enhance profitability, "all things being equal". With regards to household size, the farmers have a mean or an average of eight (8), which indicate on average the number of individuals in each household. The minimum and maximum household size is one (1) and 43 respectively. In relation to tractor services utilization, the larger the household size, the less the utilization rate of tractor services. On average

the number of extension visit is one (1), with minimum and maximum visits of 0 and seven (7) respectively. The more the number of extensions visits, the high the farmers ability to utilize tractor services in their maize production.

Table 3: Socio-Demographic and Production Characteristics of Respondents

Categorical Variables	Frequency	Percentage
Sex		
Male	235	65.5
Female	124	34.5
Marital Status		
Never married	72	20.1
Married	250	69.9
Divorced	29	8.1
Widowed	8	2.2
Religion		
Christianity	235	65.5
Islam	105	29.2
Traditionalist	19	5.3
Maize production as main occupation		
Yes	318	88.6
No	41	11.4
Other occupation		
Salary works	39	10.9
Trader	137	38.2
Private business	183	51.0
Acquisition of farmland		
Personal land/family	150	41.8
Lease	191	53.2
Inherited	12	3.3
Communal land	6	1.7
Member of a farmers' association		
Yes	145	40.4
No	214	59.6
Use of tractor services		
Yes	322	89.7
No	37	10.3

Continuous variables	Mean	Std. deviation	Minimum	Maximum
Age	44.67	14.150	18	79
Household size	7.35	4.818	1	43
Years in formal education	7.93	4.334	0	20
Years in maize farming	15.91	12.647	1	55
Number of acres	5.84	5.241	1	50
Extension visits	1.40	1.297	0	7

Source: Field survey, 2024

Tractor Services Utilization

The survey revealed that 89.7% of maize farmers in Ejura-Sekyedumase, or 322 out of 359 respondents, use tractor services for both ploughing and shelling on their farms, while only 10.3%, or 37 respondents, do not. This finding is consistent with a study by Taiwo and Kumi, 2015; which confirmed that maize shelling in the study area is largely mechanized.

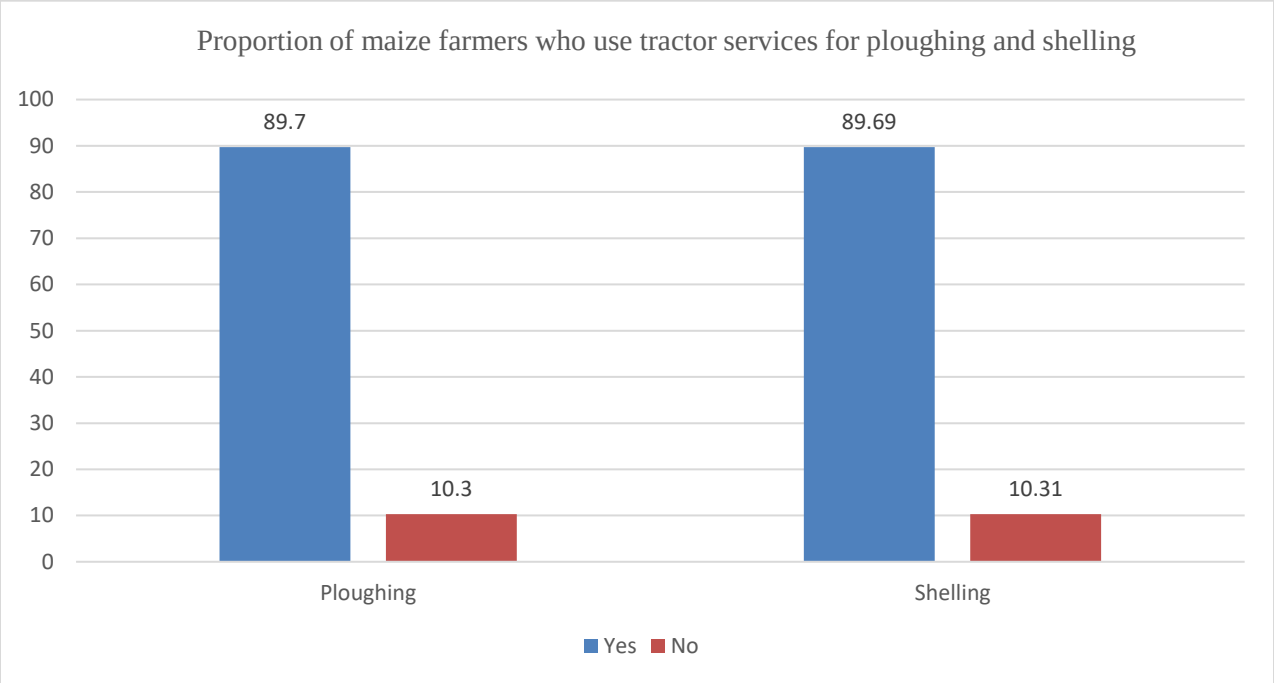

Figure 2: Proportion of Maize Farmers who use Tractor Services for Ploughing and Shelling
Source: Field survey, 2024

Farmers' Choice of Tractor Service in Maize Production

Table 4.2 indicates that maize farmers are positively influenced by farming experience, fertilizer quantity and profit per acre but negatively influenced by household size, farm size and FBO in using tractor services for shelling. The model is well fitted with a {Wald chi2(18) =104.73, p=0.0000}. The chi-square tests showed that the use of tractor services choices between the two tractor services were not jointly exclusive. A marginal effect of -0.02735 indicates that an additional member of the household, decreases the probability of using tractor services for shelling by 2.735 percentage points. One of the reasons could be that households with larger family size would perform agricultural activity by family members since they had available labor force to organize farming activity. This finding concurs with the study by Selam *et al.* (2024) that larger family size has high living standard and as a result may not rent/adopt tractor services.

A marginal effect of -0.05081 indicates that a unit increase in farm size decreases the probability of using tractor services for shelling by 5.081 percentage points. As the farm size increases, the likelihood of crop yield to increase is very high. However, tractor services decrease as more households opt for manual shelling methods, which are less efficient. This agrees with a study by Anang & Asante (2020). A marginal effect of -0.17277 indicates that maize farmer belonging to any farmer-based association, decreases the probability of using tractor services for maize shelling by 17.277 percentage points. A marginal effect of 0.01993 indicates that as the experience of the farmer increase by one-year, it increases the probability of using tractor services for shelling by 1.993 percentage points. This agrees with a study by Anang & Asante (2020) which concludes that a positive influence of farming experience on access to tractor service (shelling) is not surprising because experience increases farmers' understanding of the prevailing farming system making them more likely to appreciate the compatibility of mechanization (shelling) with existing practices A marginal effect of 0.05979 indicates that a unit increase in the quantity of fertilizer applied, increases the probability of using tractor services for shelling by 5.979 percentage points. This might be that an increase in fertilizer applied, could increase crop yield leading to an increase in the use of tractor services for shelling and hence ensuring efficiency. A marginal effect of 0.00016 indicates that as the profit per acre increase by Gh₵1.00, the probability of using tractor services for shelling increases by just 0.016 percentage points. This could

be due to the fact that maize farmers are willing to invest in tractor services through efficient and effective practices to ensure timeliness in operation in meeting demand.

Table 4 indicates that the use of tractor services for ploughing is positively influenced by farming experience, fertilizer quantity and profit per acre but negatively influenced by farm size. A marginal effect of 0.02099 indicates that as the experience of the farmer increase by one-year, it increases the probability of using tractor services for ploughing by 2.099 percentage points. This agrees with a study by Anang & Asante (2020). A marginal effect of 0.06161indicates that a unit increase in the quantity of fertilizer applied increases the probability of using tractor services for ploughing by 6.161 percentage points. According to Binswanger and Ruttan (1978) in South Asia, farmers who use tractor services produced significantly higher yields, but almost all of these cases could be attributed to increased fertilizer usage. A marginal effect of 0.00016 indicates that an increase in profit per acre by Gh₵1.00, increases the probability of using tractor services for ploughing by 0.016 percentage points. As the farm size increase by a unit, it decreases the probability of using tractor services for ploughing by 5.54 percentage points. This agrees with a study by Anang & Asante (2020).

Table 4: Multivariate Probit Model Analysis on Factors that affect maize Farmers' Choice of Tractor Services

	dy/dx	std. err.	z	P>|z|	[95% confidence interval]	
Use of tractor for shelling						
Age	0.00109	0.00503	0.22	0.829	-0.00877	0.01094
Household size	**-0.02735****	**0.01289**	**-2.12**	**0.034**	**-0.05262**	**-0.00209**
Education	0.00634	0.01120	0.57	0.571	-0.01561	0.02829
Farming experience	**0.01993*****	**0.00655**	**3.04**	**0.002**	**0.00709**	**0.03276**
Farm size	**-0.05081*****	**0.01583**	**-3.21**	**0.001**	**-0.08182**	**-0.01979**
FBO	**-0.17277***	**0.09995**	**-1.73**	**0.084**	**-0.36868**	**0.02313**
Sex	0.02163	0.09548	0.23	0.821	-0.16552	0.20878
Fertilizer quantity	**0.05979*****	**0.01343**	**4.45**	**0.000**	**0.03346**	**0.08612**
Profit acre	**0.00016*****	**0.00004**	**4.00**	**0.000**	**0.00008**	**0.00023**
Use of tractor for ploughing						
Age	-0.00086	0.00979	-0.09	0.930	-0.02005	0.01832
Household size	-0.02792	0.02105	-1.33	0.185	-0.06917	0.01333
Education	0.00404	0.02106	0.19	0.848	-0.03723	0.04531
Farming experience	**0.02099***	**0.01259**	**1.67**	**0.096**	**-0.00370**	**0.04567**
Farm size	**-0.05540***	**0.02852**	**-1.94**	**0.052**	**-0.11130**	**0.00051**
FBO	-0.20711	0.20472	-1.01	0.312	-0.60835	0.19413
Sex	0.02382	0.18966	0.13	0.900	-0.34791	0.39554
Fertilizer quantity	**0.06161****	**0.02405**	**2.56**	**0.010**	**0.01446**	**0.10875**
Profit acre	**0.00016****	**0.00007**	**2.29**	**0.026**	**0.00002**	**0.00030**

Log likelihood = -106.00424, Number of Observation=359, Wald chi2(18) = 104.73
Prob > chi2 = 0.0000; Note(s): Std. E- Standard error
dy/dx – marginal effect**,** *, **and *** indicate significant level at 10%, 5% and 1% respectively

Source: Field survey, 2024

Effect of Utilization of Tractor Services on Maize Farm Profitability in Ejura-Sekyedumase

The average income statement of farmers on a per-acre basis was used in determining the total revenue, total cost, gross profit, and net profit (Table 5). These performance indicators were used to compare those farmers who use tractor services and those who did not use tractor services.

From the survey, it was established that 322 out of the 359 respondents representing 89.7% used tractor services on their maize farm. This shows that most of the maize farmers in Ejura-Sekyedumase use tractor services.

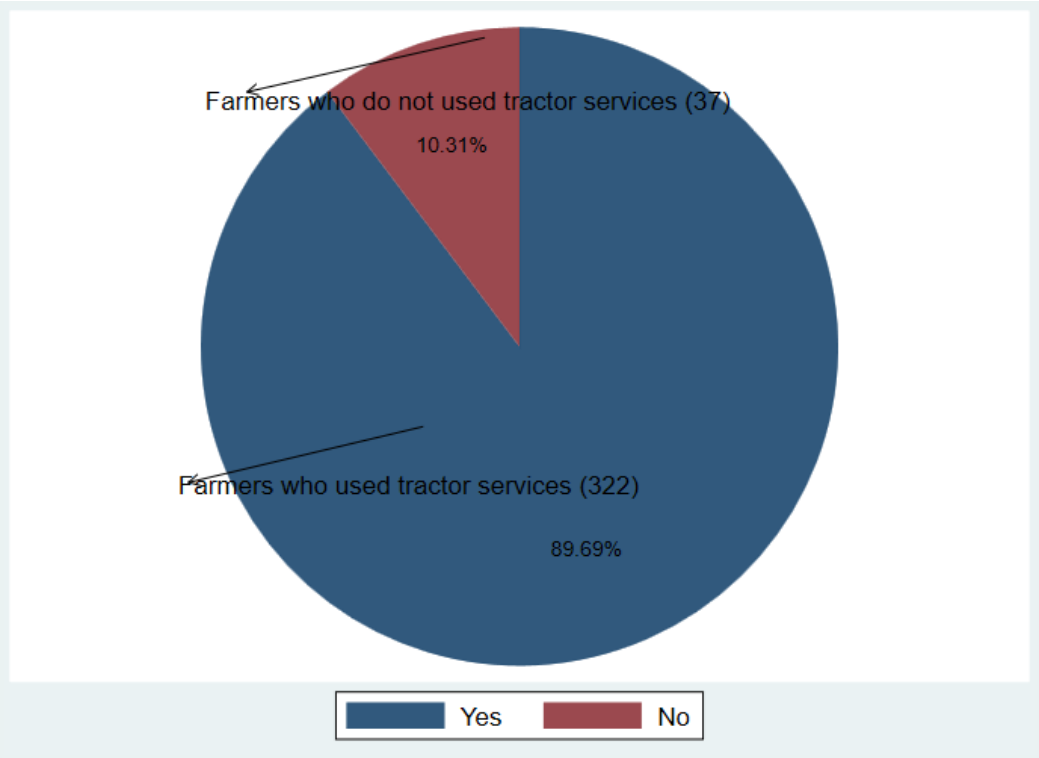

Figure 4: Farmers' use of Tractor Services
Source: Field survey, 2024

Null Hypothesis (H_0): The use of tractor services has no significant effect on maize farm profitability between the two groups. **Alternate Hypothesis (H_1)**: The use of tractor services has a significant positive effect on maize farm profitability between the two groups. The results show that the **p-value** for the t-test is **0.0131,** which is less than the significance level of 0.05. Therefore, we **reject** the null hypothesis and conclude that there is a significant **difference statistically** between the mean net profit of the two groups. With this, it shows that the usage of tractor services has a significant positive effect on the profitability of maize farmers in Ejura-Sekyedumase municipality.

Table 5: Comparison of Profit Per Acre: Tractor Service Users vs Non-Users (Based on 50kg Bags)

		Use tractor services (322)			Farmers who do not use tractor services (37)		
Variables	**Measurements**	**Qty.**	**Price (₵)**	**Cost (₵)**	**Qty.**	**Price (₵)**	**Cost(₵)**
Revenue(**a**)	50kg bag	22	250	**5500**	16	250	**4000**
Variable Costs							
Seeds	12kg	1	82	82	1	82	82
Cost of fertilizer	25kg	3	344	1032	2	344	688
Labour Cost	No. of labors	2	210	420	3	210	630
Pesticides	(in Liters)	3	60	180	2	60	120
Weedicides	(in Liters)	1	103	103	2	103	206
Storage Cost	(No. of 50kg bags)	22	20	440	16	20	320
Transportation	(Number of 50kg bags)	23	25	575	18	25	450
Shelling	(Number of 50kg bags)	23	20	460			
Total Variable Cost (b)				**3292**			**2496**
Gross Margin (c=a-b)				**2208.00**			**1504.00**
Fixed Costs							
Land cost		1	80	80	1	80	80
Land clearing		1	250	250	1	150	150
Cutlass cost		1	35	35	2	35	70
Spraying Machine		1	200	200	1	200	200
Total Fixed Cost (d)				**565**			**500**
Net Profit (a-(b+d))				**1643.00**			**1004.00**

Group	**Obs**	**Mean**	**Std. err.**	**Std. dev.**	**[95% confidence. interval]**	
No	37	597.6882	198.1922	1205.556	195.7359	999.6406
Yes	322	1193.141	77.65594	1393.486	1040.362	1345.92
Combined	359	1131.771	73.13381	1385.688	987.9454	1275.597
Diff		-595.4529	238.8042		-1065.093	-125.813

diff = mean (No) - mean (Yes), t = -2.4935, Degrees of freedom = 357

H_0: diff = 0,

H_1: diff < 0 | **H_1: diff! = 0** | H_1: diff > 0

Pr (T < t) = 0.0066 | **Pr (|T| > |t|) = 0.0131**** | Pr(T > t) = 0.9934

Obs=Observation Std. err. = Standard Error, Std dev.=Standard Deviation

Source: Field survey, 2024

From the field survey, the findings indicated that the gross profit margin of the farmers who used tractor services stood at 40.15% and those who did not used tractor services stood at 37.6% indicating that farmers who used tractor services had a higher gross profit margin. Even though farmers who used tractor services have higher operating cost, these are been covered by the benefits of using tractor services (e.g., better land preparation) making them to save about 2.55% more than those who do not used tractor services. The use of tractor services significantly impacts net profit margins, with users achieving a 29.87% margin compared to non-users at 25.1%. This represents a 4.77 percentage point advantage for farmers who utilize tractor services, highlighting the financial benefits of adopting this technology. A tractor service margin of 12.91%, implies that for every revenue generated, farmers who used tractor services spent 12.91% of their revenue as cost on tractor services (ploughing and shelling cost). Farmers who use tractor services has a return to cost of 43% and those who did not use tractor services has a return to cost of 34%, which indicates a 9% more return spent on cost for those who use tractor services.

Table 6: Performance Indicator

Indicator	Farmers who use Tractor Services	Farmers who do not
Gross Profit Margin	(2208/5500) × 100=**40.15%**	(1504/4000)×100=**37.6%**
Net Profit Margin	(1643/5500) × 100=**29.87%**	(1004/4000) × 100= **25.1%**
Return to Cost	(1643/3857) × 100=**43%**	(1004/2996)×100=**34%**
Tractor Margin	(710/5500) × 100=**12.91%**	

Source: Field Survey, 2024

Constraints in Accessing Tractor Services

Table 7 shows the constraints maize farmers face in accessing and utilizing tractor services. Financial issues were found to be the most pressing challenge maize farmers in Ejura-Sekyedumase face in accessing tractor services for farming with a mean score of 1.73 from the various challenges identified through literature and confirmed by our respondents. Environmental regulations or concerns was established as the least most pressing challenge with a mean score of 5.48. The Kendall's coefficient of concordance (W) for the rankings of the constraints faced by the maize farmers was **38.1%,** this indicates that there is a moderate agreement among the various respondents in ranking the seven (7) constraints. The asymptotic distribution was significant at **1%.**

For farmers that have access to tractor services, their most pressing challenge was financial issues with a mean rank of 1.73, and environment concern was the least pressing challenge with a mean rank of 5.59. The Kendall's coefficient of concordance (W) of the constraints faced by maize farmers who have access to tractor services was 41.4%; this indicates that the agreement among the various respondents in ranking the seven constraints is moderate. The asymptotic distribution was significant at 1%. With farmers that do not have access to tractor services, financial issues are their most pressing challenge with a mean rank of 1.72, while environmental concern is the least challenging with a mean rank of 4.54. The Kendall's coefficient of concordance (W) for the rankings of the constraints faced by maize farmers who do not have access to tractor services was **57.1%,** this shows that the agreement among the various respondents in ranking the seven constraints is good. The asymptotic distribution was significant at 1**%.**

Table 7: Constraints' maize farmers face in accessing tractor services for farming

No.	Constraints	Yes		No		All	
		Mean Rank	Rank	Mean Rank	Rank	Mean Rank	Rank
1	Financial Issues	1.73	1st	1.72	1st	1.73	1st
2	Machines not available	3.37	2nd	6.97	7th	3.34	2nd
3	Lack of infrastructure	3.43	3rd	3.36	3rd	3.42	3rd
4	Small farm size	4.78	5th	3.82	4th	4.68	5th
5	Cultural factors (e.g. taboo)	5.55	6th	4.53	5th	5.44	6th
6	Environmental Concern	5.59	7th	4.54	6th	5.48	7th
7	Lack of government support	3.56	4th	3.05	2nd	3.91	4th
	N	322		37		359	
	Kendall's W	0.414		0.571		0.381	
	Chi-Square	799.800		126.731		820.932	
	Df	6		6		6	
	Asymp. Sig.	0.000		0.000		0.000	

Source: Field survey, 2024

Summary and Conclusion

The study assessed the utilization of tractor services on maize farm profitability in the Ejura-Sekyedumase Municipality. A comprehensive survey was administered to 359 maize farmers, utilizing semi-structured questionnaires and personal interviews to gather insights into their socio-economic backgrounds and general farm characteristics, which were subsequently analyzed using descriptive and inferential statistics. Multivariate result showed that farming experience, fertilizer quantity and profit per acre has positive influence on tractor service use for shelling, while household size, farm size and FBO has a negative influence. Farming experience, fertilizer quantity, and profit per acre positively influence tractor service use for ploughing, while farm size has a negative influence.

The gross profit margin and net profit margin were used to assess the effect of tractor services on the profit of the maize farmers. The net profit per acre was compared between those who use tractor services and those who do not use tractor services on their maize farm. The result from the t-test showed a significant difference between the profit of the farmers who use tractor services and those who do not use tractor services. Based on this it was concluded from the survey that, using tractor services is quiet enough to guarantee that maize farmer will be profitable, however this profit is associated with higher costs of operation. Finally, Kendall's coefficient of concordance was used to assess the agreement among the constraints farmers face in accessing and utilizing tractor services on their maize farms. Financial issues were found to be the most pressing constraints of the farmers. Environmental regulations or concerns was found to be the least pressing constraint with a Kendall's W of **38.1%** which indicates a moderate agreement among farmers as to the constraints they face.

Policy Recommendations for Effective and Efficient utilization of tractor Services

The following suggestions were made based on the conclusions drawn from the study to further hasten tractor services utilization rate in order to overcome potential hindrances leading to effectiveness and efficiency and hence enhanced profitability to ensure sustainability.

1. Conduct Regular Impact Assessments: Implement a system for ongoing evaluation of tractor service utilization and its impact on maize farming profitability, allowing for data-driven adjustments to policies and programs.

2. Improve Access to Credit: Facilitate financing options for smallholder farmers, especially women, through microfinance institutions or government-backed loans focused on agricultural mechanization (tractor services).

3. Enhancing information symmetry: By increasing the number of extension officers, farmers can receive the necessary support and information to improve their farming techniques and overall output effectively. Offering training and extension services to educate farmers on effective tractor use and best maize farming practices, enhancing profitability.

4. Promote Women's Participation: Policymakers must create supportive opportunities and incentives that encourage women to engage in maize production through tractor service use, which can enhance food security and foster rural development. By empowering women in agriculture, we can promote practices to improve the livelihoods of families and communities.

Limitations of The Study and Suggestions for Future Research

The study only considers maize farmers' perspectives, potentially overlooking insights from tractor service providers, agricultural experts, and policymakers that could provide a more comprehensive understanding. Future research should include insights from a broader range of stakeholders, such as tractor service providers, agricultural experts, and policymakers to improve upon the tractor services utilization rate. The study does not account for climatic and environmental conditions, which are crucial factors influencing maize farm profitability.

Future research should investigate the impact of these climatic and environmental variables alongside the utilization of tractor services to provide a more comprehensive understanding of their effects on maize farming outcomes. This approach will help identify how different conditions interact with tractor services and inform strategies to optimize productivity in varying environments. It is also necessary to use balanced scorecard method to better access the performance of the maize farmers in the use of tractor services for farming. This method employs both financial and non-financial measures of performance.

Acknowledgment

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References

Abdulai, S., Nkegbe, P., & Donkoh, S. A. (2013). Technical efficiency of maize production in northern Ghana. *African Journal of Agricultural Research*, *8*(43), 5251–5259.

Aikins, K. A., Afriyie, J. K., Amcanor, I. N., Ackah, S. M., & Bobobee, E. Y. H. (2016). Assessment of tractor maintenance practices of tractor operators at Ejura, Ghana. *International Journal of Science and Engineering Applications*, *5*(5), 257–267.

Akramov, K., & Malek, M. (2012). *Analyzing profitability of maize, rice, and soybean production in Ghana: Results of PAM and DEA analysis* (GSSP Working Paper No. 0028). International Food Policy Research Institute.

Amankwah, A. (2021). *Adoption of multiple agricultural technologies and impact on productivity in rural Nigeria – A plot-level analysis*. http://ageconsearch.umn.edu

Anang, B. T., & Asante, B. O. (2020). Farm household access to agricultural services in northern Ghana. *Heliyon*, *6*(11), e05517. https://doi.org/10.1016/j.heliyon.2020.e05517

Alston, J. M., & Pardey, P. G. (2014). Agriculture in the global economy. *Journal of Economic Perspectives*, *28*(1), 121–146.

Baah, C. (2014). *Assessment of the youth in agriculture programme in Ejura-Sekyedumase district* [Master's thesis, Kwame Nkrumah University of Science and Technology]. http://ir.knust.edu.gh/bitstream/U23456789/7052/U/BAAH%20CHARLES.pdf

Beatrice Bempomaa (2014). Technical efficiency in maize production in Ejura-Sekyedumase district.

Binswanger, H. P., & Ruttan, V. W. (1978). *Induced innovation: Technology, institutions, and development*. Johns Hopkins University Press.

Chitheka, J. K., Setiadi, A., & Mariyono, J. (2024). T-test analysis for income comparison between members and non-members of Livunzu Cooperative in Chikwawa district, Malawi. *IOP Conference Series: Earth and Environmental Science*, *1364*(1), 012030. https://doi.org/10.1088/1755-1315/1364/1/012030

Codjoe, G. P., Antwi-Agyei, P., & Gyampoh, B. A. (2021). The effect of climate variability on maize production in Ejura-Sekyedumase, Ghana. *African Geographical Review*, *40*(3), 256–269.

Dagninet, A., & Wolelaw, E. (2016). Agricultural mechanization: Assessment of mechanization impact experiences on the rural population and the implications for Ethiopian smallholders. *Engineering and Applied Sciences*, *1*(2), 39–48.

De Brauw, A., & Bulte, E. (2021). *African farmers, value chains and agricultural development: An economic and institutional perspective*. Palgrave Macmillan.

Frances, C., Nazaire, H., & Collins, A. A. (2016). *Development of agricultural mechanization in Ghana: Network actors and institutions – A case study of Ejura-Sekyedumase district*.

Hoki, M., Horrio, H., & Singn, G. (1992). *Agricultural engineering literacy in developing countries*. Cornell University Press.

Jansen, A. J. (1969). Social implications of farm mechanization: Final report on cross-national research. *Sociologia Ruralis*, *9*(4), 340–407.

Kepner, R. A., Bainer, R., & Barger, E. L. (1978). *Principles of farm machinery* (3rd ed.). AVI Publishing Company.

Mdoda, G., Mdletshe, S. T. C., Dyiki, M. C., & Gidi, L. (2022). The impact of agricultural mechanization on smallholder agricultural productivity: Evidence from the Mnquma Local Municipality in the Eastern Cape province. *Mechanization and Agricultural Intensification*, *12*(3), 44–57.

MoFA. (2007). *Food and agriculture sector development policy (FASDEP II)*. Ministry of Food and Agriculture.

Fofanah, M. L. (2020). Investigating the impact of mechanized farming on the success of smallholder farmers to enhance and promote sustainable food production in Sierra Leone: An empirical case study. *International Journal of Development and Sustainability*, *9*(2), 135–150.

Musungwini, S. (2018). Mobile phone use by Zimbabwean smallholder farmers: A baseline study. *The African Journal of Information and Communication*, *22*, 29–52. https://doi.org/10.23962/10539/26171

Nariswari, T. N., & Nugraha, N. M. (2020). Profit growth: Impact of net profit margin, gross profit margin and total assets turnover. *International Journal of Finance & Banking Studies*, *9*(4), 87–96. https://doi.org/10.20525/ijfbs.v9i4.937

Obour, P. B., Arthur, I. K., & Owusu, K. (2022). The 2020 maize production failure in Ghana: A case study of Ejura-Sekyedumase municipality. *Sustainability*, *14*(6), 3514. https://doi.org/10.3390/su14063514

Perkins, T. A., Scott, T. W., Le Menach, A., & Smith, D. L. (2013). Heterogeneity, mixing, and the spatial scales of mosquito-borne pathogen transmission. *PLOS Computational Biology*, *9*(12), e1003327. https://doi.org/10.1371/journal.pcbi.1003327

Pingali, P., Bigot, Y., & Binswanger, H. P. (1987). *Agricultural mechanization and the evolution of farming systems in sub-Saharan Africa*. World Bank and CABI.

Pingali, P. (2007). Agricultural growth and economic development: A view through the globalization lens. *Agricultural Economics*, *37*(S1), 1–12. https://doi.org/10.1111/j.1574-0862.2007.00231.x

Reid, J. F., Norris, W. R., & Schueller, J. (2003). Reducing the manufacturing and management costs of tractors and agricultural equipment. *Agricultural Engineering International: CIGR Journal*, *5*, 1–12.

SARI. (1996). *Savannah Agricultural Research Institute: Annual report 1996*. Tamale: CSIR-SARI.

Selam, T., Berhanu, K., & Amsalu, B. (2024). Determinant of adoption of agricultural machine renting in West Gojjam zone, Ethiopia. *Cogent Food & Agriculture*, *10*(1), 2380123. https://doi.org/10.1080/23311932.2024.2380123

Sihlobo, W., Kapuya, T., & Baskaran, G. (2021). African perspectives and global insights: Sub-Saharan Africa's agriculture and COVID-19. *Policy Insights*, *105*, 1–8.

Taiwo, A., & Kumi, F. (2015). Status of agricultural mechanization in Ghana: A case study of maize producing farmers in Ejura/Sekyedumase district, Ashanti Region. *International Research Journal of Engineering and Technology*, *2*(6), 1282–1287. www.irjet.net

Yohanna, J. K. (2004). A survey of tractors and implements utilization for crop production in Nasarawa State. *Proceedings of the 5th International Conference of the Nigerian Institution of Agricultural Engineers (NIAE)*, *26*, 53–55.